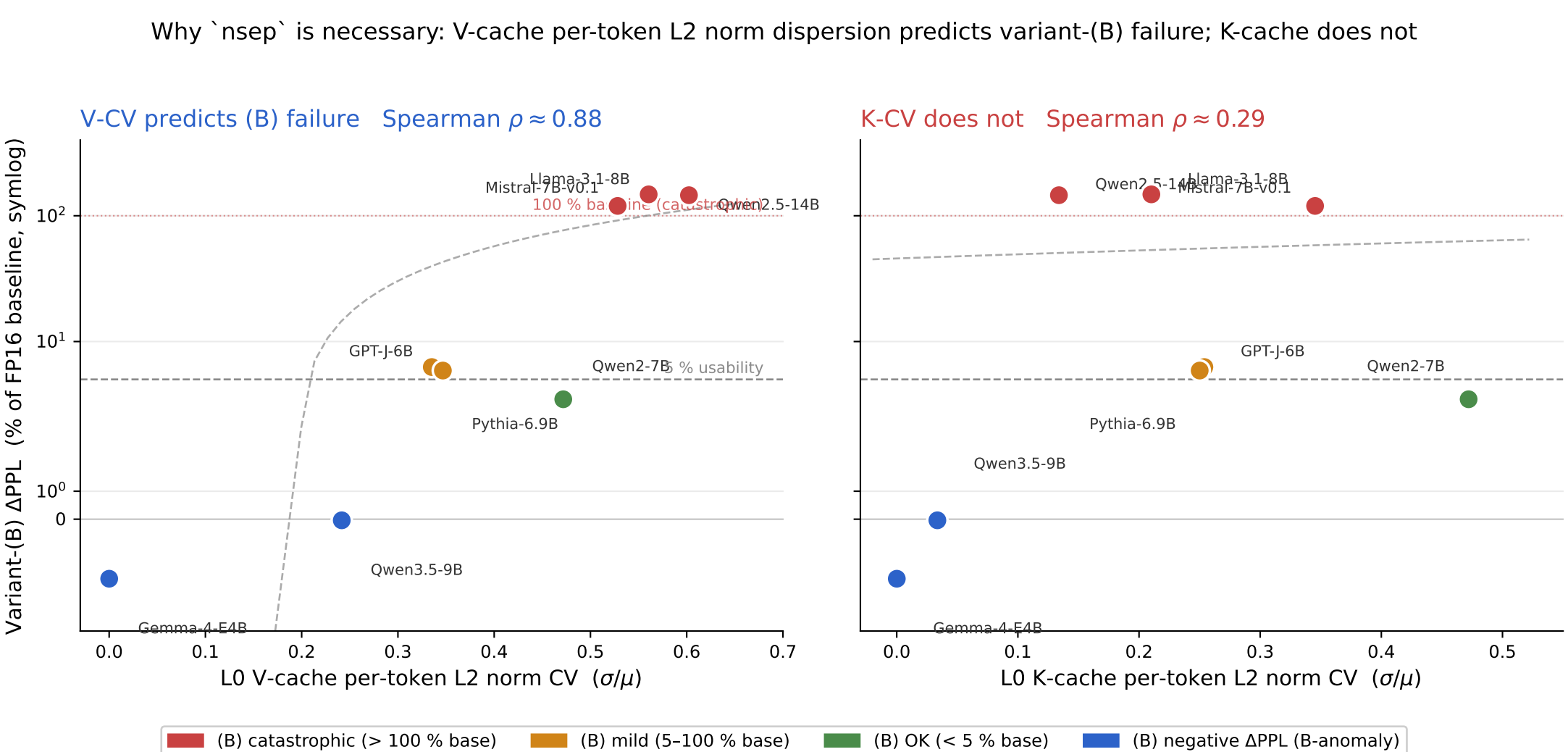
Why `nsep` is necessary: V-cache per-token L2 norm dispersion predicts variant-(B) failure; K-cache does not
V-CV predicts (B) failure Spearman ρ ≈ 0.88
K-CV does not Spearman ρ ≈ 0.29
Variant-(B) ΔPPL (% of FP16 baseline, symlog)
L0 V-cache per-token L2 norm CV (σ/μ)
L0 K-cache per-token L2 norm CV (σ/μ)
Llama-3.1-8B
Mistral-7B-v0.1
Qwen2.5-14B
GPT-J-6B
Qwen2-7B
Pythia-6.9B
Qwen3.5-9B
Gemma-4-E4B
5 % usability
(B) catastrophic (> 100 % base)
(B) mild (5–100 % base)
(B) OK (< 5 % base)
(B) negative ΔPPL (B-anomaly)